

Herald



Herald — Continuation

Field	Value
Revision	24
Created	2026-05-20
Last modified	2026-06-02
Status	active
Status summary	r25 (2026-06-02) — Messaging-features day: Slack live parity + threading + thread-context, all LIVE-PROVEN, all pushed (Herald HEAD 14fed75, 4 mirrors verified via ls-remote; docs_chain 02eb81b origin; all submodules clean). Delivered this session (9 Herald commits 005ff22→14fed75 + docs_chain 80d3d8d/02eb81b): (1) docs_chain fully integrated — engine Phase-4/5 gaps filled (watch daemon, md[?]sqlite, colorize-html), latent row-hash bug fixed, real-binary e2e GREEN; Herald 66-doc corpus drift caught + reconciled, <code>verify herald-docs exit 0</code>; routine doc flow now <code>docs_chain sync herald-docs (not export_docs.sh)</code>. (2) Slack live parity — send + inbound Socket-Mode + reply-delivery + §11.4.104 resolver + <code>--channels</code> flag (1f2b60c/43abe9b/f5c9aa5); evidence <code>docs/qa/HRD-115-LIVE-*</code>. (3) Threading both channels — Slack <code>thread_ts</code>, Telegram <code>reply_to_message_id</code>, via channel-agnostic <code>extractReplyToID</code> (28af644/e412383); live-proven. (4) Inbound threaded messages — a Subscriber replying WITHIN a thread is answered IN the same thread; live-proven. (5) Thread-context-awareness (daebeffc/bc246a2) — <code>InboundEvent.ThreadContext</code>; adapters gather the thread (Slack <code>conversations.replies</code>; Telegram quoted-parent, Bot-API single-parent limit documented); envelope conveys WHO (participants human/bot) + prior messages + WHAT (<code>detectWorkableRefs</code> HRD-N/ATM-N → names existing item, else classify newly-reported-issue/existing-item/system-event); LIVE-PROVEN BOTH channels via an <code>HERALD_CLAUDE_BIN</code> envelope-capture wrapper (the <code>cc.dispatch JOURNAL</code> does NOT contain the rendered envelope — only raw user message — so the wrapper captures <code>claude --print argv</code>). (6) Real bug fixed (14fed75) — fork/exec deadlock under concurrent dispatch (Setpgid child wedges fork→execve → parent blocks in forkExec pipe-read; stress test hung 10min under -race); fix = <code>forkExecLock</code> across <code>cmd.Start()</code> only; -race -count=3 stress PASS. (7) <code>mtproto.Client.SendReply</code> added (quoted reply). Worked subagent-driven, max parallelism, file-DISJOINT (2 batches/7 agents); conductor = integration+commit+push seam; live tests serialized (shared channel/chat); anti-bluff real evidence, zero secrets in git (verified).

Field

Value

OPEN follow-ups (honest, candidate parallel streams for tomorrow):

HRD-159 (CC fresh-session empty-reply, cross-channel — Tier-2 reply not delivered; only Tier-1 deterministic + autonomy chain proven); Slack participant-handle resolution (raw user-ids not @usernames, app-token QA user shows as bot, inbound sender maps to channel-id not poster); Slack live LEG-2 occasional flakiness under heavy channel load (re-run passes); HRD-115 ready for Issues→Fixed; broader open HRDs (081, 131, 150/155/156/157/158 ATMOSphere cluster — do cohesively, 008). Live-test cheat-sheet + gotchas in `.remember/remember.md` (local). Prior r22 (2026-05-29 day-4) — ATMOSphere²Herald workable-items integration DELIVERED end-to-end across 3 repos (Herald 2412512 / HelixConstitution 8ec2b38 / ATMOSphere 5462a38; all synced).** Phase 1 plan ~/`Documents/ATMOSphere_Herald_Integration_Plan.md` (+PDF/HTML/DOCX) from 5 parallel verified-source analysis agents. Phase 2 Herald build (all `-race` anti-bluff-tested, pushed to 4 mirrors, commits 92f0136→a541acc): HRD-148 `commons_workable` (pure-Go SQLite SSoT store+CRUD+ATMOSphere-## §... [ATM-NNN]-parser+per-property diff change-feed), HRD-149 `commons_watch` (fsnotify+WAL-poll+debounce), HRD-151 `pherald/internal/workflow` change→CloudEvent→Runner bridge + Jira/ClickUp diff formatter, HRD-152 `inbound.item.update/item.delete/investigation.start + ItemMutator + act-with-confirmation` CONFIRM flow (wired into `pherald listen --db`), HRD-153 `pherald watch` end-to-end daemon (real-SQLite+fsnotify+diff+dispatch e2e), HRD-154 preference/quiet-hours routing (TZ-aware), e2e invariant **E139**, operator guide docs/guides/WORKABLE_ITEMS_INTEGRATION.md. Phase 3 deployment: HelixConstitution workable-items parser extended for ATMOSphere format backward-compatibly (995bd2f, 4 upstreams) + anti-bluff covenant QWEN.md leading-clause for cross-file consistency (8ec2b38, 4 upstreams); on the ATMOSphere host docs/workable_items.db SSoT MATERIALIZED (58 items, validate OK, byte-identical round-trip, §11.4.95 version-controlled, d41dda5) + tools/herald registered as a PROPER submodule @ Herald main 80cb734 replacing the broken orphan gitlink (3368645) + constitution pointer bumps (ecdec56/5462a38) — pushed to github(ATMOSphere1234321)+vasic-digital-mirror (gitlab is a structural split-archive mirror, left to operator split-tooling per §11.4.41). Issues.md r39→r42; HRD-148..158 filed/advanced. Prior r21 (still valid history):** **2026-05-29 — Session-resume sweep: constitution sync c640947→3244bf4 + §11.4.100 propagation + 3 cleanup commits (HEAD 873aab4; constitution at 3244bf4; all 4 mirrors synced).** Resumed work after operator power-off. (a) **Constitution sync** — pulled HelixConstitution c640947→3244bf4 (gained §11.4.100 Video color + visual-quality fidelity mandate). (b) **§11.4.100 propagation** to Herald root (commit 5aece5d) — CLAUDE.md r15→r16, AGENTS.md r11→r12, QWEN.md, docs/guides/HERALD_CONSTITUTION.md r9→r10 with new §108.o; Herald-applicability classification: **non-applicable-but-cite** (Herald has NO video-playback surface — pherald downloads video attachments as opaque sha256-blobs via HRD-005 attachment pipeline, never decodes/renders; mandate binds latently if Herald ever ships a video-rendering surface); cascade-parallel to §108.k AOSP-pattern; literal anchor 11.4.100 present in all 4 docs per upcoming CM-COVENANT-114-100-PROPAGATION pre-build gate. (c) **§107.y anti-bluff fix** (commit c0903bc) — purged bin/qaherald (32MB compiled binary auto-committed by JetBrains IDE in commit 9796506) + closed .gitignore gap (bin/ + .test + herald-.exe/bin); root cause documented: IDE Auto-commit mechanism (also caused historical 1ce7cda PDF

Field

Value

sweep + b2d4b18 rename sweep); 32MB blob remains in git history at 9796506 — purging requires git-filter-repo + force-push which §11.4.41 gates on operator authorization (filed as accepted tech-debt). (d) **§11.4.19 atomic Issues→Fixed migration of HRD-139..142** (commit 873aab4) — Wave 8 Track B closure recorded in Fixed.md r16→r17: HRD-139 (MTProto Client live wiring), HRD-140 (Subscribe roundtrip), HRD-141 (Wave 6 closed loop), HRD-142 (Wave 6.5 lifecycle) all graduate directly from TaskList → Fixed.md. (e) **§11.4.90 Obsolete graduation of 3 legacy NON-COMPLIANT tests** (same commit 873aab4) — HRD-143 marks TestSubscribe_LiveBotAPI Obsolete (superseded by §11.4.98+HRD-140; literal source-comment "requires the operator to hand-send a message" cited per §11.4.6 no-guessing); HRD-144 marks tests/test_wave6_live_loop.sh Obsolete (superseded by §11.4.98+HRD-141); HRD-145 marks qaherald lifecycle --manual mode Obsolete (superseded by §11.4.98+HRD-142). Each Obsolete-Details cites Since/Reason/Superseding-item + triple-check evidence per §11.4.6. (f) **Docs sibling-export audit** — 117 fresh / 0 stale .md [?] .html/.pdf/.docx pairs across the tracked tree (Python-based mtime-comparison) — Herald's scripts/export_docs.sh discipline holds. Inheritance gate 15 PASS / 0 FAIL post-§11.4.100 propagation. 3 commits c0903bc + 5aece5d + 873aab4 all pushed to 4 mirrors via §11.4.88 background-push (per-mirror flock, push detached after commit-lock release). Prior r20 (still valid history):** **2026-05-28 (day-3) — Wave 8 Track B COMPLETE + v0.6.0 tagged (HEAD b313a5e; constitution at 15cd4bc; all 4 mirrors synced)**. Wave 8 Track B (MTProto user-account harness for §11.4.98 full-automation testing) is now end-to-end live-evidence-proven: HRD-139 qaherald/internal/mtproto/ Client live wiring (238448e — gotd/td v0.144.0 vendored as a submodule, Connect/SendMessage/WaitForReply/WhoAmI/Close interface, 4198-byte encrypted session file at ~/.config/herald/mtproto.session, mode 0600) + qaherald mtproto login/whoami/logout Cobra subcommands (ec4c3c0 — added HERALD_MTPROTO_ALLOW_NON_TTY=1 env override for delegated-by-conductor login per the operator "can you run it alone?" mandate; conductor-driven FIFO-piped OOB-code flow proven autonomous end-to-end with operator's single 5-digit chat-paste). HRD-140 TestMTProto_Subscribe_AutonomousRoundTrip PASSED LIVE in 1.30s (367d58b — MTProto sends a stimulus + polls bot getUpdates for delivery confirmation, replacing the legacy TestSubscribe_LiveBotAPI Bot-API-only path which is fundamentally non-autonomous per the §11.4.98 contract). HRD-141 TestMTProto_Wave6_AutonomousClosedLoop PASSED LIVE in 48.58s (7e50ed0 then re-fixed in d7d6524 + 0797e85 — spawns pherald listen with cmd.Dir = repoRoot (CRITICAL — pherald resolves docs/Issues.md relative to its CWD), pre-acks pending updates via consumePendingUpdates(t, botToken) to prevent TLS-connection-reset storms from stale chat messages, asserts on the pherald --qa-out-dir JSONL journal (tgram.message in + cc.dispatch + cc.reply directions, matched on the unique test-text string rather than message_id — MTProto's int64 message_id is a different namespace from Bot API's chat-local message_id) — replaces tests/test_wave6_live_loop.sh). HRD-142 TestMTProto_Wave65_LifecycleAutonomous PASSED LIVE in 26.04s with 3/3 subtests PASS (1f2e75d — 3 representative Wave 6.5 fast-path scenarios (Help / Status / Continue), each assertion is tgram.message in + `cc.reply

Field	Value
Issues summary	9 open, all evidence-gated (Issues.md r48): HRD-008 (operator quickstart validation), HRD-081 (containers submodule upstream), HRD-115 (Slack live — provide creds per §1a), HRD-131 (SQLite SSoT — constitution-repo binary), HRD-150/155/156/157/158 (ATMOSphere — needs the repo cloned locally per §1a). The bulk of the catalogue migrated to Fixed.md. see Issues.md. r19 (Tgram-completeness wave COMPLETE) atomically migrated HRD-133/134/135/136/137/138 Issues→Fixed per §11.4.19 — see Issues.md r35 + Fixed.md r16. r18: v1.0.0 Batch A migrated HRD-018 / HRD-026 / HRD-027 Issues→Fixed (§11.4.19 atomic). v1.0.0 Batch B (HRD-019..025, §42 flavor constitution bindings) is in_progress — each flavor's internal/bindings/ Pipeline + rule catalogue + PURE detectors + emit-through-audit foundation + §107.x evidence + §11.4.85 stress all landed (f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a), but live operation-interception (the Subjects the detectors classify) is deferred to Batch C §43 command bodies, so these stay open not Fixed. Wave 7 closed HRD-110/111/112/113/114; HRD-115..121 are the remaining Wave 7 tasks T6-T12. HRD-122..HRD-130 (§11.4.85 stress+chaos suite, GAP-3) + HRD-132 (Runner idempotency-window bug) MIGRATED to Fixed.md. HRD-131 is §11.4.93 SQLite workable-items SSoT migration Phase 1 — OPEN-deferred per operator 2026-05-27: Phase 2 adoption blocked on the unimplemented + schema-incompatible constitution migration binary (assessment 36fa90b); a §11.4.74 constitution-repo task, not a Herald reimplementation.
Fixed	HRD-001..HRD-018 (less open), HRD-026, HRD-027, HRD-028, HRD-080, HRD-092..HRD-101, HRD-110..HRD-114, HRD-122..HRD-130, HRD-132, HRD-133..HRD-138
Fixed summary	see Fixed.md
Continuation	►► RESUME (next session, after 2026-05-31). Current state in §1 Snapshot; exactly-what-to-provide in §1a "TOMORROW — mandatory data to provide". HEAD bff48a5; all 4 mirrors synced; all submodules + the docs_chain sibling repo clean+pushed; gate 15/15; 9 open HRDs, all evidence-gated. To unblock: (A) drop Slack creds into .env (HERALD_SLACK_BOT_TOKEN/_CHANNEL_ID[_APP_TOKEN]) per docs/guides/messengers/SLACK.md → I run TestSlack_Live_Send/E127, capture docs/qa/HRD-115-LIVE-*, close HRD-115; (B) clone the ATMOSphere repo to ~/Projects/ATMOSphere (absent) → I reconcile HRD-155/157 drift + advance HRD-156 T-B..T-F. MTPProto session is live (E135 re-confirmed 2026-05-31, message_id=211685); docs/guides/MTPROTO.md is the re-bootstrap guide. The prior-session "ATMOSphere resume tomorrow" directive below is HISTORICAL — most of it landed (see Fixed.md + Issues.md r48). ——— (prior continuation history below) ——— ►► RESUME TOMORROW (2026-05-30) — ATMOSphere integration: LIVE EVIDENCE + DAEMON LAUNCH. STATE: code is DEPLOYED end-to-end (Herald 2412512 / HelixConstitution 8ec2b38 / ATMOSphere 5462a38); the SQLite SSoT is materialized (58 items at docs/workable_items.db on the nezha host milosvasic@10.6.100.146: /run/media/milosvasic/DATA4TB/Projects/Android_15); tools/herald submodule is live. Every Herald module + both flows (watch→diff→notify, inbound→CRUD) are hermetically anti-bluff-tested; what's NOT yet proven is the LIVE Telegram round-trip + a running daemon. NEXT STEPS (dependency-ordered): (1) HRD-156 LIVE evidence — operator runs qaherald mtpproto login ONCE (enter the 5-digit Telegram OOB code; the one step Telegram can't automate). Then conductor

Field

Value

drives the live tests: mutate the ATMOSphere SSoT → assert a REAL Telegram message reaches a subscriber (T-A); operator message → real SSoT row mutation (T-B); exact per-property-diff payload byte-assert (T-C); §11.4.85 stress/chaos (T-D/T-E); paired §1.1 mutation gate (T-F). Capture under docs/qa/HRD-156-LIVE-<TS>/ (HRD-133 sanitizer guarantees no token leak), flip the SKIP-with-reason invariants to PASS. **(2) Launch the daemon on the ATMOSphere host** — pherald watch --db docs/workable_items.db (outbound SSoT→notify) + pherald listen --db docs/workable_items.db (inbound→CRUD), wired as a host-side runner mirroring tools/oom-protect's install pattern; seed Subscribers + the main-group alias. **(3) gitlab ATMOSphere mirror** — vasic-digital/ATMOSphere-Android-15 gitlab is a STRUCTURAL split-archive mirror (1497-commit divergence + a Split-archive batch commit) — needs the operator's split-push tooling, NOT a force-push (§11.4.41). **(4) Follow-ups:** Fixed.md item-structuring (currently raw-preserved/lossless but 0 structured items — extend the constitution tool's parseFixed); per-submodule QWEN.md leading-clause cascade across the 25 ATMOSphere submodules (the covenant CORE + operative rule are ALREADY everywhere — only the one leading clause; low priority); HRD-150/regenerator productionization; HRD-154 PG-adaptor decode of subscribers.metadata.preferences (deferred to HRD-156). All integration HRDs (148-158) tracked in Issues.md r42; master plan ~/Documents/ATMOSphere_Herald_Integration_Plan.md. ——— (prior continuation history below) ——— **Resume — live-evidence collection for tgram (HEAD 77da6eb):** The tgram-completeness wave is COMPLETE in code + tests + docs (HRD-133..138 Fixed). Hermetic coverage is full (tgram suite 26/26 PASS, 0 SKIP; chaos test deterministic). What remains is the LIVE-EVIDENCE layer — currently SKIP-with-reason in scripts/e2e_bluff_hunt.sh for E17/E18 (live Telegram send + roundtrip), E34 (live Claude Code session resolution), E63-E70 (Wave 6 closed-loop with real Telegram + Claude Code), E71-E80 (Wave 6.5 ticket lifecycle wire-byte). **Operator-action required:** export real credentials in the conductor's session via ! export HERALD_TGRAM_BOT_TOKEN=<token> HERALD_TGRAM_CHAT_ID=<chat-id> HERALD_TGRAM_BOT_USERNAME=<username> HERALD_CLAUDE_BIN=<path-or-omit-for-PATH>. Per §107.x: ALL live evidence lands under docs/qa/HRD-NNN-LIVE-<TS>/ with the HRD-133 sanitizer guarantee that no token text appears in any committed transcript/log. Once operator exports, conductor drives live runs + captures evidence + flips the SKIP-with-reason invariants to PASS in the next commit. **If operator declines / defers:** next priorities are (a) Wave 8 supervised channel restart (currently fail-loud-tears-down-siblings per pherald/cmd/pherald/listen.go:387-398 doc-comment), (b) docs audit task #147 (comprehensive per-flavor operator guides), (c) HRD-131 §11.4.93 SQLite SSoT — remains OPEN-deferred until constitution-repo migration binary lands. **Prior r18 — v1.0.0 readiness (workstream 3) per plan docs/superpowers/plans/2026-05-27-v1.0.0-execution.md; Batch A + Batch B DONE, next is Batch C: (1) Batch C — §43 command catalogue HRD-029..056 (IMMEDIATE NEXT UNIT)** — the live CLI command bodies that PRODUCE the Subjects the Batch-B detectors classify (pherald commit-push, sherald destructive-guard, cherald creds-scan, rherald tag-mirror, bherald evidence-capture, etc.), grouped into 5 flavor clusters per the v1.0.0 plan. Closing Batch C is what flips HRD-019..025 from in_progress→Fixed (their bindings have no live Subjects until the command surface lands). **NOTE — the e2e invariant counter**

Field**Value**

is at E96; next free is E97+. (2) **Batch D — commons_infra HRD-081 + HRD-085..090.** (3) **Batch E — Wave 7 T6-T12 Slack adapter HRD-115..121** (vendored slack-go submodule — **verify the I6 inheritance-gate check after the .gitmodules entry**; confirm slack-go v0.16.0 field names `inner.Files/socketmode.EventTypeEventsAPI/UploadFileV2Parameters`; then qa Herald Slack MessengerClient, spec §11.0/§32.2/§43, e2e, mutation gate, Issues→Fixed/v0.6.0/4-mirror push; T5 pre-staged `perChannelConfig("slack") + blank-import in listen.go` — `HERALD_CHANNELS=tgram, slack` currently errors `ErrUnknownChannel` until T6 registers slack, by design; plan docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md line 980). **PLAN GOTCHAS:** Wave 7 e2e invariants **MUST** renumber off E81-E89 — use **E90+** (E89 is now consumed by Batch A's audit write-through); the slack-go submodule addition requires re-running the I6 inheritance-gate check (`tests/test_constitution_inheritance.sh`). Tag cadence: v0.5→v1.0.0 across the batches. (5) **GAP-4 docs/qa back-fill + docs-audit #147** — features lacking docs/qa/<run-id>/evidence per the ee63ce6 qa-coverage audit + comprehensive documentation audit. **DEFERRED / GATED ITEMS: §11.4.93 SQLite SSoT migration (HRD-131) — OPERATOR-DEFERRED 2026-05-27:** Phase 2 adoption is blocked on the constitution's workable-items migration binary, which is an unimplemented scaffold (0/7 subcommands) + schema-incompatible with Herald's HRD-NNN pipe-table format; per §11.4.74 Herald may NOT reimplement it — the fix is a constitution-repo task affecting all consumers (assessment docs/research/workable-items-phase2-assessment-2026-05-27.md, 36fa90b). HRD-131 stays OPEN, blocked-on-constitution-binary. MTPROTO real-channel automation — **OPERATOR-DECISION:** provide my.telegram.org app_id+app_hash (folds into a qa Herald MessengerClient MTPROTO impl reusing the entire scenario engine; Telegram never relays bot-to-bot in groups — see docs/research/telegram-bot-to-bot-constraint.md); operator should also /revoke @pherald_qa_bot (leaked plaintext token). §11.4.90 obsolescence — zero current Obsolete rows; convention in place, periodic re-audit pending. (RESOLVED this session (2026-05-28 day-2): v1.0.0 Batch B COMPLETE — all 7 §42 flavor constitution bindings HRD-019..025 landed (f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a), each internal/bindings/ Pipeline + rule catalogue + PURE detectors + §107.x evidence + §11.4.85 stress 1200-concurrent, e2e E90-E96 wired (HRD-019..025 stay in_progress — live-op interception is Batch C); constitution pulled 3c9c4e9→15cd4bc + §11.4.95-97 propagated (d5b9895, gate 15/0); 3 fixes — #194 audit-test qaRoot t.TempDir guard (a8ad4f7), eventbus PublishedByType data race → LoadOrStore (f5d1367, a real E3 -race regression latent since Batch A), E17 e2e SKIP-guard now requires live PG (75e7a49). Prior r13: v1.0.0 Batch A — HRD-018/026/027 closed atomically Issues→Fixed, e2e **81 PASS / 0 FAIL** (E89 wired), commons_constitution audit write-through §107 bluff FIXED + admin mode-flip REST; conductor-verified -race -count=3 + real-PG + §11.4.85 stress 512-row no-lost-writes. Earlier: GAP-3 §11.4.85 stress+chaos suite IMPLEMENTED — e2e 78 PASS / 0 FAIL — AND atomically migrated Issues→Fixed (HRD-122..130 Completed); HRD-132 Runner idempotency-window bug FIXED 354b883 (claim-before-dispatch, exactly-once sends==1) + migrated to Fixed.md; constitution §11.4.89-94 propagation + residue scanner; HRD-123 flaky oversized-body assertion + E82 stale-anchor anti-bluff issues fixed. Prior r10 RESOLVED: Wave 6.5 mutation-gate M3 stale anchor; Postgres-SASL e2e FAILs; HRD-101 filed as a real Fixed.md row.) §11.4.87 endless-loop: continue until Issues.md zero-active +

Field	Value
	CONTINUATION §3 empty + no subagent in flight. NO BLUFF. Push 4 mirrors every 2-3 commits.

Table of contents

- [§0. How to use this document](#)
- [§1. Snapshot](#)
- [§1a. ★ TOMORROW — mandatory data to provide](#)
- [§2. Last commit landed](#)
- [§3. Active work](#)
- [§4. Next concrete steps](#)
- [§5. Long-form pointers](#)

§0. How to use this document

Paste the following block into any CLI agent (Claude Code / OpenCode / Cursor / Aider / Gemini CLI) to resume Herald work exactly where it was left:

You are working on the Herald project at ~/Projects/Herald (also reachable as the Herald/ submodule of a consuming project). The Helix Universal Constitution lives at <ancestor>/constitution/ (parent-walk discovery). Read in this order: CLAUDE.md, AGENTS.md, README.md, docs/guides/HERALD_CONSTITUTION.md, docs/guides/CONSTITUTION_INHERITANCE.md, docs/specs/mvp/specification.V4.md. Then read docs/CONTINUATION.md (this file) for live state, docs/Issues.md for open work, docs/Status.md for current phase, docs/Fixed.md for closed history. Go workspace builds via go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/.... Inheritance gate tests/test_constitution_inheritance.sh MUST exit 0 before any commit. Multi-mirror fan-out push to four hosts (GitHub + GitLab + GitFlic + GitVerse) is mandatory per Constitution §103.

§1. Snapshot

- **HEAD:** 14fed75 (2026-06-02). **All 4 mirrors (GitHub / GitLab / GitFlic / GitVerse) in sync** (verified via ls-remote); all 9 Helix-stack submodules + containers/ clean + at pinned commits; the docs_chain sibling repo (~/Projects/docs_chain) clean + pushed at 02eb81b.
- **Active spec:** docs/specs/mvp/specification.V4.md (V1–V3 archived under docs/specs/mvp/archive/).
- **Workspace: 18 Go modules** (10 shared/foundation + 8 flavor binaries). Live + tested: pherald_listen inbound runtime, Telegram + Slack adapters, Claude Code dispatch, commons_storage (pgx + River + Redis), the Gin /v1/* REST API. **Every module now has an operator/developer guide** under docs/guides/ (the 8 shared-module guides landed 2026-05-31).
- **Latest tag: v0.6.0** (2026-05-28).
- **Inheritance gate:** 15 PASS / 0 FAIL.
- **e2e_bluff_hunt: 167 PASS / 0 FAIL / 0 stress-chaos-evidence SKIP** (last full run); invariants through **E147** (E140-E142 participant-attribution, E143-E145 intent-recognition, E146 docs_chain pilot drift-gate, E147 docs_chain fleet drift-gate). Next free id **E148+**.

- **Docs Chain:** integrated end-to-end (engine fix + §11.4.106 governance + 66-doc fleet, verify exit 0); binary built from ~/Projects/docs_chain → /tmp/docs_chain.
- **MTPROTO live harness:** session active at ~/.config/herald/mtproto.session (@milos85vasic); E135 round-trip re-confirmed LIVE 2026-05-31 (message_id=211685).
- **Open HRDs: 9, all evidence-gated** — HRD-008, 081, 115, 131, 150, 155, 156, 157, 158 (see docs/Issues.md r48). The non-gated autonomous queue is drained.

§1a. 🌟 TOMORROW — mandatory data to provide (then I resume closing the gated tracks)

The non-gated autonomous queue is **drained**; all 9 open HRDs are evidence-gated. To unblock, provide ONE or both:

A — Slack live credentials → I close HRD-115

Add to .env (git-ignored — never commit it), then tell me "Slack creds are in .env":

```
# === SLACK (HRD-115) ===
HERALD_SLACK_BOT_TOKEN=xoxb-<bot token>      # api.slack.com/apps
    -> your app -> OAuth & Permissions -> Install to
    Workspace -> copy "Bot User OAuth Token"
HERALD_SLACK_CHANNEL_ID=C<channel id>        # target channel ->
    View channel details -> bottom -> Channel ID; then invite
    the bot to that channel
HERALD_SLACK_APP_TOKEN=xapp-<app token>      # ONLY if you want
    inbound (Socket Mode) replies; omit for outbound-only
```

Full step-by-step (create app, scopes, install, Socket Mode, channel ID, the live-test command): docs/guides/messengers/SLACK.md. Once provided I run `go test -tags=integration -count=1 -run TestSlack_Live_Send ./qaherald/internal/messenger/...`, capture docs/qa/HRD-115-LIVE-<TS>/, and migrate HRD-115 Issues→Fixed.

B — Clone the ATMOSphere repo → I advance HRD-156 / 150 / 155 / 157 / 158

The ATMOSphere repo is **not present locally** (~/Projects/ATMOSphere absent), so the ATMOSphere tracks cannot be reconciled or advanced from here. Clone it alongside Herald (`git clone <url> ~/Projects/ATMOSphere`) and tell me — I'll reconcile the HRD-155/157 status-vs-narrative drift against real upstream state + advance HRD-156's remaining T-B..T-F test layers (the MTPROTO live foundation is already proven, E135 PASS 2026-05-31).

Already ready (no action needed tonight)

- **MTPROTO:** session live + verified (qaherald mtproto whoami OK as @milos85vasic); docs/guides/MTPROTO.md is the complete re-bootstrap guide if the session ever expires.
- **HRD-008** (quickstart): validate the §26.5 compose stack on your hardware per docs/guides/OPERATOR_CREDENTIALS.md when convenient.

The §4 "Next concrete steps" further below is from a prior session and is **superseded** by this §1a block + the §1 snapshot.

§2. Last commit landed

HEAD bff48a5 (2026-05-31). This session landed, all pushed to 4 mirrors: **Docs Chain integration end-to-end** (CLI binary-hash verify fix + §11.4.106 governance propagation into all 4 root docs + pilot + **66-doc fleet** with E146/E147 drift-gates); **4 parallel streams** (Obsolete-HRD migration, docs_chain consumer docs, open-work audit, project-status currency); **HRD-117 closed** (Issues 10→9); **Status.md / README / HERALD_CONSTITUTION §101 / CLAUDE.md (70XXX→24XXX ports)** refreshed to current reality; spec-V4 path-sync; **2 credential guides** (messengers/SLACK.md + new MTPROTO.md, §11.4.99-verified vs official docs); **8 shared-module guides** (all 18 modules now have guides); GitHub secret-scan false-positive resolved (operator unblock) → 4-mirror sync; **MTPROTO live re-confirmed** (E135 PASS, message_id=211685). Prior-session history below. —————

HEAD 534e785 (2026-05-30) — test(§11.4.98): gate/retire 4 NON-COMPLIANT attended tests behind opt-in skips + (prior, same session) HRD-146 dispatch hardening a975d8b, per-flavor guides 87878fb, api-bank wiring c80ad81, docs-audit 338a896, stress/chaos ee5566f, CLI-bank shell: bce6de7. The session incorporated HelixQA as Herald's anti-bluff QA SSoT end-to-end (see §3). helixqa pushed to its 3 mirrors at 9e0ec3b; HelixConstitution Fixed.md parser at 9b74f96. **Prior** 873aab4 — docs(§11.4.19 + §11.4.90): atomic HRD-139..142 Issues→Fixed + Obsolete graduation of 3 legacy NON-COMPLIANT tests. Fixed.md r17 records HRD-139/140/141/142 (Wave 8 Track B) as Fixed; Issues.md r36 marks 3 legacy tests Obsolete per §11.4.90 with full Obsolete-Details + triple-check evidence per §11.4.6. **Prior** 5aece5d — feat(§11.4.100): propagate Video color + visual-quality fidelity mandate (non-applicable-but-cite for Herald). Constitution sync c640947→3244bf4; §11.4.100 propagated to all 4 root docs; inheritance gate 15/0. **Prior** c0903bc — fix(anti-bluff): purge bin/qaherald 32MB binary + close .gitignore gap. §107.y working-tree quiescence fix — purged JetBrains-IDE-Auto-committed 32MB compiled binary from index + sealed the gitignore gap. **Prior** b313a5e — docs(qa): Wave 8 Track B live evidence — E135/E136/E137 all PASS LIVE under §11.4.98. Wave 8 Track B (MTPROTO user-account harness for §11.4.98 full-automation) CLOSED end-to-end: HRD-139 MTPROTO Client + qaherald mtpROTO login/whoami/logout (238448e + ec4c3c0); HRD-140 Subscribe roundtrip PASSED LIVE in 1.30s (367d58b); HRD-141 Wave 6 closed loop PASSED LIVE in 48.58s with journal-based §11.4.98 autonomy assertion (7e50ed0 + d7d6524 + 0797e85); HRD-142 Wave 6.5 lifecycle PASSED LIVE in 26.04s, 3/3 subtests (1f2e75d); E135-E137 wired into bluff hunt (a80d206); HelixConstitution §11.4.99 anchored upstream + Herald root + 9 Helix-stack submodules (fb9d81e + d1100b2 + dc007ab + d1b6f25); evidence committed under docs/qa/HRD-LIVE-MTPROTO-20260528T125321Z/ (b313a5e). v0.6.0 ANNOTATED TAG cut. **Prior** 77da6eb — feat(tgram HRD-137 + Wave C): §11.4.85 stress+chaos SKIP→PASS + 6 HRDs Issues→Fixed. **Tgram-completeness wave COMPLETE — HRD-133..138 atomic Issues→Fixed per §11.4.19.** 9 files changed (3 modified + 6 created): a one-line commons_messaging/channels/tgram/subscribe.go seam fix (threads a.baseURL into telebot.Settings.URL before NewBot, parallel to ensureBot at tgram.go:60-67; production path unchanged when a.baseURL == ""); new commons_messaging/channels/tgram/stress_chaos_test.go carrying TestTgram_Stress_MultiRecipientFanOut (already PASS pre-fix, Wave-B-era) + TestTgram_Chaos_GetUpdatesPollerResilience (httptest fault-injector — 3× 500 + 2× 1.5s hangs honoring ctx-cancel + 1× mid-body Hijacker connection-close + 2× success);

docs/qa/HRD-137-20260528T150000Z/stress_chaos/ artefacts (anchor preserved verbatim, transcript §2 documents SKIP→PASS, chaos_assertion.txt rewritten with measured PASS-state values); atomic Issues.md r34→r35 + Fixed.md r15→r16 migrating all 6 tgram-completeness HRDs (HRD-133/134/135/136/137/138) Issues→Fixed. Deterministic across -race -count=3 — every iteration identical: recovery=6 fault(500=12 hang=8 close=3 ok=6) goroutines 3→4 (leaked=1) duration=12.07s. Full tgram suite **26/26 PASS, 0 SKIP** (was 24/24 with chaos SKIPped). All 4 mirrors synced.

§3. Active work

2026-06-02 — Messaging features (Slack live parity + threading + thread-context)

COMPLETE + live-proven + pushed. NOTHING IN FLIGHT. Clean checkpoint: Herald 14fed75 on all 4 mirrors (ls-remote verified), docs_chain 02eb81b origin, all submodules clean. The day's work (docs_chain integration, Slack live parity, threading both channels, inbound-threaded processing, thread-context-awareness with participants+subject, fork/exec deadlock fix, mtproto SendReply) is summarized in the r25 Status-summary above with commit ids; live evidence under docs/qa/HRD-115-LIVE-* + docs/qa/HRD-thread-context-tgram-*. Worked subagent-driven (2 batches/7 agents, file-disjoint, conductor-integrated). **OPEN follow-ups (next session — candidate parallel streams):** (1) **HRD-159** CC fresh-session empty-reply (cross-channel — Tier-2 CC reply not delivered; touches commons_messaging/dispatch/claude_code). (2) **Slack participant-handle resolution** — thread-context Participants show raw Slack user-ids not @usernames, the app-associated QA user shows as "bot", and the Slack inbound sender maps to the channel-id not the poster (touches commons_messaging/channels/slack/ + the §11.4.104 resolver). (3) **Slack live LEG-2 flakiness** under heavy same-channel load (re-run passes — consider adaptive timeout / channel cleanup). (4) **HRD-115** is round-trip+threading+context live-proven → ready for Issues→Fixed (§11.4.19). (5) Broader open HRDs (081, 131, 150/155/156/157/158 ATMOSphere cluster — cohesive not split, 008). Live-test cheat-sheet + gotchas in .remember/remember.md (local-only handoff buffer).

2026-05-30 — HelixQA incorporated as Herald's anti-bluff QA SSoT (COMPLETE, real-evidence-proven). Nothing in flight. HelixQA (git@github.com:HelixDevelopment/helixqa.git) is a sibling repo at ~/Projects/helixqa (CONST-051 flat layout) with all 7 own-org deps at ~/Projects/ (snake_case per CONST-052, operator-approved). Herald drives it via scripts/helixqa_run.sh + banks under challenges/helixqa-banks/. Delivered this session (Herald e36929a→534e785, helixqa c0b2757→9e0ec3b, all pushed):

- **Both Herald banks genuinely execute, mutation-proven** — CLI bank (10 cases, 8 flavor binaries) + api bank (11 /v1/* cases) rewritten to self-asserting shell:/curl actions (HelixQA run executes shell: on the desktop path; assertions bite — corrupting one flips PASSED→FAILED). Evidence: docs/qa/HRD-QA-CLI-FLAVORS-*/, docs/qa/HRD-API-BANK-WIRING-*/.
- **5 real anti-bluff bugs found + fixed in helixqa** (it's the org-wide SSoT): i18n printf echo; desktop crash-absence promotion bluff (+ generalized to all platforms — a SKIP with any unrun asserting step is never promoted); §107.x captured-evidence reporter (qa-report.md now carries the real command + exit code); autonomous executor shell: wiring (was filing false-negative tickets); dead swipe: action + description: verdict divergence.
- **LLM/Vision bridge** = HelixQA's existing BridgedCLIProvider (auto-discovers claude on PATH) — zero new bridge code; helixqa autonomous runs end-to-end with it (\$0 free providers).
- **Herald-side hardening landed alongside:** §11.4.85 stress/chaos for commons_workable + Notifier (ee5566f); §11.4.98 full-automation — 4 attended tests gated behind opt-in,

MTPProto replacements confirmed (534e785); **HRD-146 dispatch hardening** — 120s timeout + process-group kill on the `claude_code` dispatcher (real 30s-shim killed in 702ms); 8 per-flavor operator guides + §11.4.99 footers (87878fb); docs-audit corrections (module/flavor/migration counts → 18/8/000014, honest DB-materialization status — 338a896).

Cross-repo also landed: HelixConstitution Fixed.md tracker parser (9b74f96, 4 mirrors); ATMOSphere constitution pointer bump (643b36c); HRD-158 QWEN anti-bluff-clause cascade reconciled across 18/19 ATMOSphere submodule repos; codegraph re-indexed (5042 files, validator 7/0).

Remaining (next session): (a) the **~52-HRD Issues→Fixed closure migration** — the production-readiness map (a read-only audit this session) found ~52 HRDs are code-complete + tested + evidence-backed (E90-E96 e2e invariants VERIFIED present) but stuck in `open/in_progress` because the §11.4.19 atomic migration was never run; this is the v1.0.0 critical path and it is paperwork, not engineering; (b) HRD-147 (governance dead_letter subscriber — genuinely not-started, needs design); (c) the 1 skipped QWEN repo `tools/helixqa/HelixQA` (has an unrelated 2nd commit — operator decision); (d) CONST-052 REMOTE GitHub repo renames (deferred — cross-org, GitHub redirect); (e) HRD-131 §11.4.93 SQLite SSoT (constitution-repo binary still unimplemented).

§4. Next concrete steps

1. **► v1.0.0 critical path — the ~52-HRD Issues→Fixed closure migration (paperwork, not engineering).** A read-only production-readiness audit (2026-05-30) found ~52 HRDs are code-complete, tested, and evidence-backed but sitting in `open/in_progress` because the §11.4.19 atomic migration was never run. The "after Telegram works" gate (HRD-156) is ALREADY live-proven, and the §43 binding e2e invariants E90-E96 are VERIFIED present in `scripts/e2e_bluff_hunt.sh`. Work: per closeable HRD, atomically move its entry `Issues.md`→`Fixed.md`, bump the revision metadata, regen siblings. Closeable-now (no operator): HRD-019..025 (flavor bindings, E90-E96), HRD-029..056 (§43 commands), HRD-085..089 (pgx repo ops), HRD-116/118/119/120/121, HRD-148/149/151/152/153/154 (ATMOSphere WS — table is stale vs the r42 header which already says "landed"), HRD-146 (now hardened+tested this session). Anti-bluff: verify each HRD's real test/qa evidence before migrating (don't trust the self-reported status). Genuinely not-done (skip): HRD-147 (dead_letter subscriber, needs design). Externally-gated (skip): HRD-115-live/157/150/155/131/158/008/081.
2. **Live evidence — Telegram + Claude Code closed-loop** (gated on operator export). Drive `tests/test_wave6_live_loop.sh` end-to-end with real bot + chat. Capture transcript under `docs/qa/HRD-100-LIVE-20260528-<TS>/`. Flip E17/E18/E34/E63-E70 SKIP→PASS in `scripts/e2e_bluff_hunt.sh`. Per §107.x: ALL live evidence lands with the HRD-133 sanitizer guarantee (no token text in any committed transcript/log).
3. **Live evidence — Wave 6.5 ticket lifecycle** (same credentials). Drive the qaherald lifecycle scenarios against the real bot. Capture under `docs/qa/HRD-NNN-LIVE-W6.5-<TS>/`. Flip E71-E80 SKIP→PASS.
4. **Tag v0.6.0** (after E17-E80 live evidence + Slack live operator-provided OR explicit operator deferral). Wave 7 T6-T12 close at the same boundary; HRD-115..121 `Issues`→`Fixed`.
5. **Tag v0.7.0 / v1.0.0-dev-NN cadence** — operator-provided live credentials unblock the v0.6→v1.0 chain (mirrors the v0.4.0 / v0.5.0 deferred-until-live-evidence pattern).
6. **Deferred / gated items:** §11.4.93 SQLite SSoT (HRD-131, constitution-repo task), MTPProto real-channel automation (operator decision), Wave 8 supervised channel restart (currently fail-loud-tears-down-siblings per `pherald/cmd/pherald/`

listen.go:387-398 doc-comment), docs audit task #147 (comprehensive per-flavor operator guides).

§4a. Wave 6 live-test handoff (T10b — gating tag v0.4.0)

Operator-supplied credentials required — Wave 6's e2e invariants E63-E70 land as honest SKIPs until the live closed-loop runs against real Telegram + real Claude Code with real chat messages exchanged.

Step-by-step.

1. Export credentials in the shell that will drive the test:

```
export HERALD_TGRAM_BOT_TOKEN="<your-bot-token-from-  
BotFather>"  
export HERALD_TGRAM_CHAT_ID="<numeric-chat-id-where-bot-is-  
admin>"  
export HERALD_CLAUDE_BIN="$(command -v claude)" # path to  
the claude CLI binary  
export HERALD_PROJECT_NAME="Herald" # or any  
operator-chosen session name
```

2. Pick a <run-id> (timestamp + 4-char nonce):

```
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-$(uuidgen | cut -c1-4)"  
QA_DIR="docs/qa/HRD-100-${RUN_ID}"  
mkdir -p "${QA_DIR}/attachments"
```

3. Type ONE message in the configured Telegram chat (script reads it via getUpdates within 60s).
4. Run the live closed-loop test:

```
bash tests/test_wave6_live_loop.sh
```

The script builds pherald, observes the original message_id, starts pherald listen --bot-token "\$HERALD_TGRAM_BOT_TOKEN" --chat-id "\$HERALD_TGRAM_CHAT_ID" --qa-out-dir "\$QA_DIR" in the background, waits up to 45s for a reply with reply_to_message_id == original, and exits 0 on PASS.

5. Capture supplementary logs:

```
cp /tmp/pherald-w6.log "${QA_DIR}/pherald-listen.log"  
# claude stdout/stderr — paths depend on the journaling  
setup. Copy whatever  
# the test produced into ${QA_DIR}/claude-stdout.log +  
claude-stderr.log.
```

6. Author a brief **\${QA_DIR}/README.md** (5 lines minimum) narrating: who ran it, when, what message was sent, what reply came back, any anomalies. This is NOT auto-generated — operator-written narrative is the §107.x evidence anchor.

```
7. git add ${QA_DIR}/ && git commit -m "Wave 6 step 10b: docs/
   qa/HRD-100-${RUN_ID}/ live closed-loop evidence" then proceed to
   T13b (v0.4.0 tag + 4-mirror push).
```

If the script **SKIPS** (any env var unset): the test prints `SKIP: <reason>` and exits 0. That is honest §11.4.3 hardware-absent SKIP-with-reason, not a PASS. Tag `v0.4.0` **MUST NOT** be created until at least one PASS run is committed to `docs/qa/HRD-100-<run-id>/`.

§4b. qaherald-auto resume runbook (next session — gating tag v0.5.1)

Status as of 2026-05-23 session end:

- HEAD `8ad4cb9` on main; all 4 mirrors in sync (`github.com:vasic-digital/Herald.git`, `gitlab.com:vasic-digital/herald.git`, `gitflic.ru:vasic-digital/herald.git`, `gitverse.ru:vasic-digital/Herald.git`).
- Tag `v0.4.0` shipped (Wave 6 closed-loop). Tag `v0.5.0` shipped (Wave 6.5 lifecycle implementations — premature: only S1+S2 live evidence + carries the `pre-53a7ad3` envelope-action-guidance bug). Tag `v0.5.1` is the target after qaherald-auto T6-T8 land + full 15-scenario evidence committed.
- Plan: `docs/superpowers/plans/2026-05-23-qaherald-auto.md` (commit `52ce679`).
- Done: T1 skeleton + T2 messenger + T3 scenarios + T4 orchestrator + T5 report (combined commit `8ad4cb9` for T3-T5).
- TODO: **T6 pre-flight** + **T7 hermetic tests** + **T8 shell-adapter** then live run + tag.

Resume prerequisites

The following are **already provisioned** in `~/ .zshrc` (export-on-shell-start):

```
HERALD_TGRAM_BOT_TOKEN
    # @atmosphere_worker_bot — pherald-bot (Telegram bot
    8823384001)
HERALD_TGRAM_CHAT_ID      # -4946584787 (ATMOSphere Development
    group)
HERALD_OPERATOR_IDS       # 2057253161 (the operator's user-id)
HERALD_QA_BOT_TOKEN       # @pherald_qa_bot — 2nd bot
    (8971749017, privacy DISABLED, member of the group)
```

The qa-bot token is plaintext in `~/ .zshrc` — operator should `/revoke` via `@BotFather` after Wave 6.5 close-out and regenerate (the token leaked into one prior session's bash output earlier today, was never committed to git but lives in `/private/tmp/claude-501/...` operator-local files).

The containers `herald-postgres` (port 24100) and `herald-redis` (port 24200) are LIVE — both have been Up since the start of this session; data persists across podman restarts. Schema is at migration v12 (all 12 applied).

Resume sequence (suggested next-session order)

1. Pull state + verify:

```
cd ~/Projects/Herald && git pull origin main && git log --
    oneline -10
git status --short # expect clean
git stash list      # 2 stashes preserved: T8 Wave-5 salvage
    + premature S3-S5 transcript WIP
```

2. **Spawn T6 (pre-flight)** via a fresh subagent — read docs/superpowers/plans/2026-05-23-qaherald-auto.md Task 6 in full. Implementation lands at qaherald/internal/lifecycle/preflight.go + preflight_test.go. 10 gates: pherald-bot reachable via getMe, qa-bot reachable, qa-bot privacy disabled, qa-bot is group member, OPERATOR_IDS contains qa-bot, etc. Each gate has distinct exit code for diagnostics.
3. **Spawn T7 (full hermetic tests)** — extends T3-T5's 12 tests with httptest-based scenario simulation. Coverage targets: every scenario PASS path + every FAIL diagnostic + S9 SKIP path + S14 outbound-attachment sha256 round-trip + S11/S12/S13 inbound-attachment download + Issues.md/Fixed.md fs-mutation assertion.
4. **Spawn T8 (shell-script adapter)** — rewrites tests/test_wave6.5_lifecycle.sh to delegate to qaherald lifecycle for automated runs. Keeps the original --manual flag for the operator-typing interactive UX as a fallback.
5. **Run the live e2e (the §107 watershed):**

```
cd ~/Projects/Herald
# Add qa-bot user-id to OPERATOR_IDS so S5/S6/S8/S10 succeed
export HERALD_OPERATOR_IDS="$
    {HERALD_OPERATOR_IDS},8971749017"
# Re-export from .zshrc (or just open a new shell)
source ~/.zshrc

# Build pherald + qaherald
go build -o /tmp/pherald ./pherald/cmd/pherald
go build -o /tmp/qaherald ./qaherald/cmd/qaherald

# Start pherald listen in background with QA journaling
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-w6.5live"
PHERALD_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`-pherald"
QAUTO_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`"
mkdir -p "${PHERALD_QA_DIR}/attachments" "${QAUTO_QA_DIR}/
    attachments"
/tmp/pherald listen --qa-out-dir "${PHERALD_QA_DIR}" --docs-
    dir docs &
PHERALD_PID=$!
trap 'kill -TERM $PHERALD_PID 2>/dev/null' EXIT
sleep 5

# Run qaherald lifecycle (T6-T8 must be done first)
/tmp/qaherald lifecycle \
    --pherald-bot-username=atmosphere_worker_bot \
```

```
--pherald-qa-out-dir="${PHERALD_QA_DIR}" \
--out="${QAUTO_QA_DIR}" \
--run-id="${RUN_ID}"
# Exit 0 on all-PASS or all-PASS-with-S9-SKIP. Non-zero on
any FAIL.
```

6. Capture evidence:

- `${PHERALD_QA_DIR}/transcript.jsonl` — pherald's view (classifications, dispatch, replies)
- `${QAUTO_QA_DIR}/transcript.jsonl` — qaherald-auto's view (sends, assertions)
- `${QAUTO_QA_DIR}/report.md` — Markdown summary
- `${QAUTO_QA_DIR}/attachments/<sha256>.<ext>` — content-addressed inbound + outbound attachments
- `docs/Issues.md` + `docs/Fixed.md` — mutated by S5/S6/S8/S10 (and reverted by S10 + S15 cleanups; both files end in valid state)

7. Author operator README + commit:

```
cat > "${QAUTO_QA_DIR}/README.md" <<EOF
# HRD-101 Wave 6.5 lifecycle evidence — <run-id>

<5+ lines narrating: who ran it, when, which scenarios
      PASSed/FAILed/SKIPped,
bot reply quality observations, any anomalies. Operator-
      written, NOT auto-generated.>
EOF
git add docs/qa/HRD-101-lifecycle-${RUN_ID}*
git commit -m "Wave 6.5 step 13: HRD-101 live 15-scenario
      lifecycle evidence (qaherald-auto)"
```

8. Tag v0.5.1 + 4-mirror push:

```
git tag -a v0.5.1 -m "Wave 6.5 close: full lifecycle live
      evidence + envelope action-guidance fix"
git push origin main
git push origin v0.5.1
# Verify all 4 mirrors converged on the same SHA + tag
for mirror in github.com:vasic-digital/Herald.git
      gitlab.com:vasic-digital/herald.git
      gitflic.ru:vasic-digital/herald.git
      gitverse.ru:vasic-digital/Herald.git; do
  git ls-remote "git@${mirror}" v0.5.1
done
```

Open / known constraints for tomorrow

- **S9 (non-operator Done: rejection):** requires `HERALD_QA_BOT_TOKEN_NON_OPERATOR` env (a THIRD bot account whose user-id is NOT in `OPERATOR_IDS`). Without it, S9 emits a `SKIP-with-reason`. Operator may

register a third bot OR accept the SKIP (S9's logic is already unit-tested hermetically in T5's mutation gate).

- **First-scenario CC bootstrap:** ~30s on first inbound message because the Claude Code session must spawn. Subsequent scenarios use the cached session (~5-15s typical).
- **Issues.md mutation race:** if you run `qaherald lifecycle` while ANOTHER pherald listen is also processing in the same checkout, Issues.md may receive double-writes. Single-pherald-listen invocation is the safe default.
- **Postgres-SASL carry-over (RESOLVED 2026-05-27):** the `e2e_bluff_hunt` FAILs on Postgres-SASL invariants (E7-E12 + E14-E16 + E37-E42) are FIXED — `9b166c4` (auth env + PG password self-heal + Redis/tenant wiring + honest SKIP-with-reason) + `bdb9f1` (Runner nil-Redis graceful PG-only fallback). At resolution `e2e_bluff_hunt` reported 60 PASS / 0 FAIL / 24 SKIP; the later GAP-3 session (`d2a60d3`) raised it to **78 PASS / 0 FAIL** with E81-E88 wired in.

Safe-shutdown checklist (HISTORICAL — end of 2026-05-23 session; superseded — current HEAD is `d2a60d3`)

Check	State
All committed work pushed to 4 mirrors	YES — HEAD <code>8ad4cb9</code> confirmed on github+gitlab+gitflic+gitverse
Local working tree clean	YES — <code>git status --short</code> empty
Stashes preserved	2 stashes: T8 Wave-5 salvage (older) + premature mid-test S3-S5 transcript WIP (newer, will be regenerated cleanly by qaherald-auto in next session)
Background pherald/qaherald processes	NONE running (<code>ps aux grep pherald</code> empty)
Container state	<code>herald-postgres + herald-redis</code> Up 17 hours — persist data across <code>podman stop && podman start</code> ; can be left running OR stopped (<code>podman stop herald-postgres herald-redis</code>); next session restart is <code>podman start herald-postgres herald-redis</code>
<code>/tmp/*.log</code> files	one operator-token leak shred-deleted earlier today; remaining transient logs contain no credentials (verified)
<code>.zshrc</code>	contains 4 <code>HERALD_*</code> exports including <code>HERALD_QA_BOT_TOKEN</code> plaintext — operator should /revoke + regenerate via @BotFather post-Wave-6.5 closure
Memory entries	All session findings persisted under <code>/Users/milosvasic/.claude/projects/-Users-milosvasic-Projects-Herald/memory/</code> — survive across sessions

Machine is safe to shut down. Resume tomorrow by `cd ~/Projects/Herald && git pull origin main` and start with `qaherald-auto T6`.

§5. Long-form pointers

- `docs/specs/mvp/specification.V4.md` — full active spec (Revision 1, supersedes V3).
- `docs/specs/mvp/specification.V4.md#30-v2-self-review-log` — every review pass.
- `docs/guides/HERALD_CONSTITUTION.md` — §101..§106 extending Universal.

- docs/guides/CONSTITUTION_INHERITANCE.md — parent-discovery + gate.
- tests/test_constitution_inheritance.sh — the gate.
- quickstart/ — HRD-008 scaffold.
- commons/types.go — the §11.0 type contract reference.